

DBMS – Seller and Shop Management

Project Title: Myntra Marketplace – Fashion and Lifestyle E-Commerce System

Skill Task: DBMS

Week: 3

1. Database Design

The 3 tables directly used in the Payment module SQL work below. Section 1.3 (Payment) is the table this report focuses on.

1.1 Customer Table

Attribute	Data Type	Constraint
Customer_ID	INT	Primary Key
First_Name	VARCHAR2(50)	NOT NULL
Last_Name	VARCHAR2(50)	NOT NULL
Email	VARCHAR2(100)	Unique Key
Phone_Number	VARCHAR2(15)	Unique
Password	VARCHAR2(50)	NOT NULL
Gender	VARCHAR2(10)	-
Address	VARCHAR2(255)	NOT NULL

1.2 Order Table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATE	-
Order_Status	VARCHAR2(20)	-
Total_Amount	NUMBER(10,2)	-

1.3 Payment Table

Attribute	Data Type	Constraint
Payment_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Payment_Method	VARCHAR2(20)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL

Payment_Date	DATE	NOT NULL
Amount	NUMBER(10,2)	NO

2. Table Relationships

Relevant entities and relationships from the Entity Analysis document, focused on the Payment module:

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Order	One-to-Many
Order	Includes	Order_Item	One-to-Many
Order_Item	Refers To	Product	Many-to-One
Order	Has	Payment	One-to-One
Order	Has	Shipment	One-to-One

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (
  2     Payment_ID      NUMBER(5) PRIMARY KEY,
  3     Order_ID        NUMBER(5),
  4     Payment_Method  VARCHAR2(20) NOT NULL,
  5     Payment_Status  VARCHAR2(20) NOT NULL,
  6     Payment_Date    DATE NOT NULL,
  7     Amount          NUMBER(10,2) NOT NULL,
  8     CONSTRAINT fk_payment_order
  9     FOREIGN KEY (Order_ID)
 10     REFERENCES Orders(Order_ID)
 11 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount)
  3 VALUES
  4 (601, 301, 'UPI', 'Successful', SYSDATE, 2999.00);
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount)
  3 VALUES
  4 (602, 302, 'Card', 'Successful', SYSDATE, 4599.00);
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount)
  3 VALUES
  4 (603, 303, 'COD', 'Pending', SYSDATE, 1899.00);
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount)
  3 VALUES
  4 (604, 304, 'Net Banking', 'Failed', SYSDATE, 3499.00);
```

1 row created.

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
601	301	UPI	Successful	16-SEP-26	2999
602	302	Card	Successful	16-SEP-26	4599

5.2 Display Failed Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
604	304	Net Banking	Failed	16-SEP-26	3499

6. Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 604;
```

1 row updated.

```
SQL> SELECT * FROM Payment WHERE Payment_ID = 604;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
604	304	Net Banking	Successful	16-SEP-26	3499

7. Analyze Payment Methods

Count of transactions by payment method:

```
SQL> SELECT
  2     Payment_Method,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
UPI	1
Card	1
COD	1
Net Banking	1

Amount collected by payment method (successful only):

```
SQL> SELECT
  2     Payment_Method,
  3     SUM(Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
UPI	2999
Card	4599
Net Banking	3499

8. Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.Customer_ID,
  5     c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  6     p.Payment_Method,
  7     p.Payment_Date,
  8     p.Amount,
```

```

9      p.Payment_Status
10 FROM Payment p
11 JOIN Orders o
12      ON p.Order_ID = o.Order_ID
13 JOIN Customer c
14      ON o.Customer_ID = c.Customer_ID
15 ORDER BY p.Payment_Date DESC;

```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PAYMENT_METHOD	AMOUNT	STATUS
604	304	404	Divya K	Net Banking	3499	Successful
603	303	403	Vikram P	COD	1899	Pending
602	302	402	Sneha M	Card	4599	Successful
601	301	401	Aakash R	UPI	2999	Successful